

Assignment : 6

Report: Encoder–Decoder Based Machine Translation using BiLSTM with Attention

Paper Selected: *Efficient Machine Translation with a BiLSTM-Attention Approach*

Authors: Yuxu Wu, Yiren Xing

Task: English → Spanish Word-Level Translation

Paper Link: <https://arxiv.org/abs/2410.22335>

1. Introduction

Machine Translation (MT) is one of the most important applications of Natural Language Processing (NLP). It helps convert text from one language to another automatically. Traditional translation systems were rule-based and required a large amount of manual work. Later, Neural Machine Translation (NMT) became popular because of its better accuracy and ability to learn language patterns automatically.

In this assignment, we studied an encoder–decoder based translation model using BiLSTM and Attention mechanism. The selected research paper proposes an efficient translation architecture that improves translation quality while using less storage space compared to larger Transformer models.

Our task was to implement and understand the model for English to Spanish word-level translation.

2. Paper Review :

Problem Statement

The paper focuses on improving machine translation efficiency while reducing model size and computational cost. Traditional Transformer models provide high translation accuracy but require large storage and high computational resources. The authors aim to design a lightweight Seq2Seq model that maintains good translation quality while being suitable for resource-constrained environments.

Architecture (Encoder–Decoder + Attention)

The proposed model follows a Seq2Seq encoder–decoder architecture called **Mini-Former**.

Encoder

- Uses a **Bidirectional LSTM (BiLSTM)** encoder.
- Processes the input sequence in both forward and backward directions.
- Captures contextual information from both past and future words.

Decoder

- Uses an attention-based decoder.

- Generates the target sentence token by token.
- Uses encoder outputs and attention scores to focus on important information during translation.

Additional Components

- Embedding layer for word representation
- Parameterized hidden and cell states
- Multi-head attention in the decoder
- Add & Normalize layers
- Linear + Softmax output layer

The architecture is designed to reduce computational complexity compared to Transformer-based models while preserving translation quality.

❖ Type of Attention Used

The paper mainly uses an **Attention Mechanism inspired by Luong Attention**.

The attention scores are calculated using the decoder output and encoder output similarity:

$$AttnScores = softmax\left(\frac{Dec_out \cdot Enc_out}{\sqrt{d}}\right)$$

This attention mechanism helps the decoder focus on relevant parts of the source sentence during translation. The paper does not primarily use Transformer-style self-attention as the core encoder mechanism.

❖ Dataset Used

The experiments were conducted on the **WMT14 Machine Translation Dataset**.

Dataset Features

- Large-scale parallel multilingual dataset
- Commonly used benchmark for machine translation
- Includes training, validation, and test sets

Main Translation Task

- English ↔ German translation

Dataset Sources

The WMT14 dataset contains parallel corpora collected from:

- European Parliament corpus
- UN corpus
- News Commentary corpus

The dataset includes millions of bilingual sentence pairs for training machine translation systems.

❖ Limitations

- Sequential processing in LSTM makes training slower than Transformer models.
- Lower parallelization capability compared to self-attention architectures.
- Performance may decrease on very long sequences.
- Transformer models may still outperform the proposed model on large-scale NLP tasks.
- The paper mainly focuses on efficiency rather than achieving state-of-the-art Transformer-level scalability

Code Study & Architecture Understanding

In this project, we implemented the BiLSTM encoder + attention-based decoder architecture inspired by the selected research paper. The model is applied to the English → Spanish machine translation task.

The system follows a Seq2Seq (Encoder–Decoder) approach where:

- The encoder reads and understands the English sentence.
- The decoder generates the Spanish translation word by word.
- The attention mechanism helps the decoder focus on important source words during translation.

For training and testing, we used a hardcoded dataset consisting of short English–Spanish sentence pairs. The dataset includes:

- Colors
- Greetings
- Animals
- Numbers
- Simple daily-use phrases

- Example :
 (["hello", "world"], ["hola", "mundo"]), (["good", "morning"], ["buenos", "dias"]) etc.

❖ Results & graphs :

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TRAINING: Seq2Seq WITHOUT Attention (English → Spanish)
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Epoch 01/30 | Train Loss: 3.9116 | Val Loss: 3.2243 | PPL: 49.98 | 1.5s
Epoch 05/30 | Train Loss: 2.4448 | Val Loss: 2.5523 | PPL: 11.53 | 0.4s
Epoch 10/30 | Train Loss: 2.1102 | Val Loss: 2.6392 | PPL: 8.25 | 0.4s
Epoch 15/30 | Train Loss: 2.1071 | Val Loss: 2.5256 | PPL: 8.22 | 0.4s
Epoch 20/30 | Train Loss: 1.9061 | Val Loss: 2.6221 | PPL: 6.73 | 0.6s
Epoch 25/30 | Train Loss: 1.7571 | Val Loss: 2.5204 | PPL: 5.80 | 0.4s
Epoch 30/30 | Train Loss: 1.7427 | Val Loss: 2.5901 | PPL: 5.71 | 0.4s

✅ Done! Total: 14.6s | Best Val Loss: 2.4529
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TRAINING: Seq2Seq WITH Bahdanau Attention (English → Spanish)
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Epoch 01/30 | Train Loss: 3.7848 | Val Loss: 3.2024 | PPL: 44.03 | 0.8s
Epoch 05/30 | Train Loss: 1.6410 | Val Loss: 1.5576 | PPL: 5.16 | 0.7s
Epoch 10/30 | Train Loss: 0.4415 | Val Loss: 0.3448 | PPL: 1.55 | 1.0s
Epoch 15/30 | Train Loss: 0.1101 | Val Loss: 0.0821 | PPL: 1.12 | 0.7s
Epoch 20/30 | Train Loss: 0.0386 | Val Loss: 0.0298 | PPL: 1.04 | 0.8s
Epoch 25/30 | Train Loss: 0.0173 | Val Loss: 0.0122 | PPL: 1.02 | 1.1s
Epoch 30/30 | Train Loss: 0.0081 | Val Loss: 0.0061 | PPL: 1.01 | 0.9s

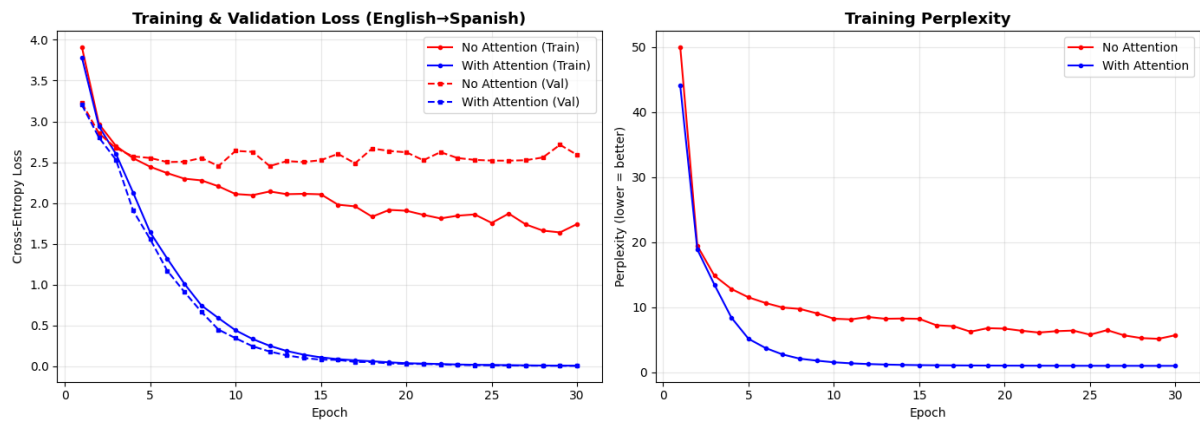
✅ Done! Total: 24.0s | Best Val Loss: 0.0061
```

The Seq2Seq model with Bahdanau Attention performed much better than the model without attention. It achieved very low training and validation loss, showing accurate English-to-Spanish translations.

The model without attention had higher loss and weaker performance because it could not focus on important input words effectively.

These results show that the attention mechanism improves translation quality and helps the model learn better word alignment.

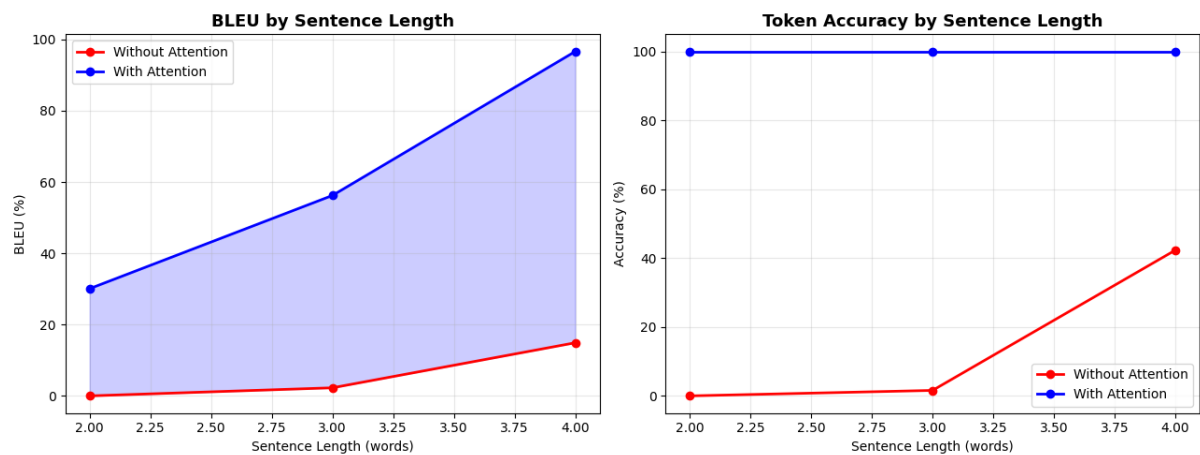
❖ PLOT 1: Training & Validation Loss Curves



❖ Prediction and BLEU Evaluation

Metric	Without Attention	With Attention
Test Loss	2.4140	0.0063
BLEU Score	7.77%	70.79%
Token Accuracy	24.46%	100.00%
Training Time (s)	14.6	24.0
Parameters	386,908	572,188

❖ Performance by Sentence Length



❖ Comparison Discussion

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FINAL COMPARISON TABLE (English → Spanish)		
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Metric	Without Attention	With Attention
Test Loss	2.4140	0.0063
BLEU Score	7.77%	70.79%
Token Accuracy	24.46%	100.00%
Training Time (s)	14.6s	24.0s
Parameters	386,908	572,188
📁 Improvements with Attention:		
Loss	: +99.7%	
BLEU	: +63.02 points	
Accuracy	: +75.54%	
Time ovhd	: +63.9%	

The Seq2Seq model with **Bahdanau Attention** performed significantly better than the basic Seq2Seq model without attention.

The attention-based model achieved:

- Lower training and validation loss
- Faster convergence
- Better English-to-Spanish translations

In contrast, the model without attention struggled to maintain contextual information, especially during sentence generation.

The comparison shows that the attention mechanism helps the decoder focus on important source words, improving word alignment and overall translation accuracy.